

RI-AutoCoder PRO — User Manual

A theory-driven qualitative coding tool. This manual walks from a single interview transcript to figures and statistics you can defend in a paper, and explains at each step **why the tool behaves the way it does** — because reviewers rarely ask which button you pressed. They ask what entitles you to the claim.

All screenshots use **synthetic demonstration data** (a fictional smart-traffic-signal case) and contain no real interview material. Some screenshots show the Chinese interface; the tool is fully bilingual and the interface language is a per-user setting.

Version: 2026-08-19 · 36 screenshots in this manual show the Chinese interface, because only part of the set was recaptured in English. Interface language is a user setting; the two are functionally identical.

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Before anything else: what this tool refuses to do

Most qualitative software treats an LLM as an autocoder: feed it a transcript, receive codes, accept them. This tool deliberately does not.

Three non-negotiable design commitments

- ① **The framework comes from literature, not from the model's memory.** A coding framework is the validity foundation of the entire study. Asking a model to recall the dimensions of a theory produces output that is plausible and partly invented. So framework construction here means: retrieve the literature, let the model summarise and draft from the retrieved abstracts, then approve it dimension by dimension yourself.
- ② **Interpretive authority stays with the researcher.** The model's coding is a *draft*. The review tab lets you confirm, revise, delete, or add segments the model missed, and every change is written to an audit trail.
- ③ **Anything computable is computed, not estimated.** Reliability comes from genuine double-blind double coding over a frame that includes unmarked units, so recall is meaningful; when the assumptions of a chi-square test fail, the tool refuses to print a p-value.

If what you want is fast automatic coding, this tool will feel obstructive. It is built so that in a viva or a review you can account for every number.

0. Installation

1. Place the folder somewhere with a path free of unusual characters.
2. Double-click `啟動_Windows.bat` (`啟動_Mac.command` on macOS, `啟動_Linux.sh` on Linux).
3. First launch builds an isolated environment and installs dependencies (3–10 minutes), then installs a Chinese word segmenter (~400 MB). Failure of the segmenter step does not prevent use.

4. A browser opens automatically. **Leave the console window open** — closing it stops the program.

Never copy the `.venv` folder. A virtual environment records absolute paths at creation time; a copy fails with `Failed to import encodings module`, which gives no hint as to the real cause. Exclude `.venv` when moving the project. If you already copied one, just run the launcher — it detects the situation and rebuilds automatically.

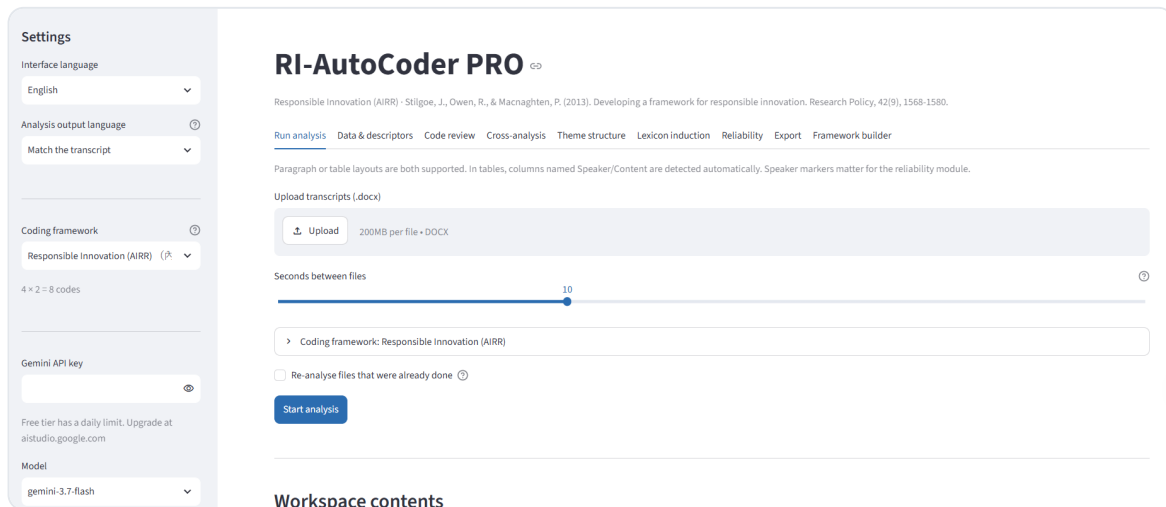


Figure 1. The nine tabs across the top are the whole workflow, and left to right is roughly the order you work in.

The sidebar

Settings

Interface language

English

Analysis output language

Match the transcript

Coding framework

Responsible Innovation (AIRR) (built

4 × 2 = 8 codes

Gemini API key

Free tier has a daily limit. Upgrade at aistudio.google.com

Model

gemini-3.7-flash

Enter an API key above to load the current model list.

Figure 2. Upper half. The two language settings are **independent**: interface language affects only labels, while analysis output language controls what language the model writes its coding rationales and theme names in. An English interface with Chinese transcripts is a legitimate combination.

The screenshot shows a web interface for configuring the OpenAlex API key. At the top, there is a section titled "OpenAlex API key" with a help icon. Below the title is a text input field containing a masked key (dots) and a toggle icon. A paragraph of text explains that setting this as an environment variable can avoid retyping. Below this is a section titled "Load existing data" with a paragraph explaining that only loaded interviews are used for analysis. There are three buttons: "All", "None", and "Rescan folder". Below these is a section titled "Interviews to load" with a dropdown menu showing "Choose options" and a "Load selected" button. At the bottom is a "Clear workspace" button.

Figure 3. Lower half: both API keys live here — Gemini for coding and theme induction, OpenAlex for literature retrieval in framework construction.

To avoid retyping the OpenAlex key, set it as the `OPENALEX_API_KEY` environment variable. It is then filled in automatically, and it does not travel with the project folder if you share the project.

1. Running an analysis

This is the only step that consumes API quota. Everything afterwards runs offline on your machine.

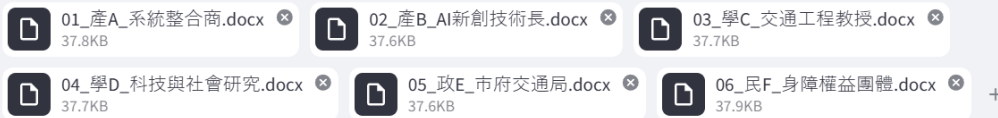
RI-AutoCoder PRO

Responsible Innovation (AIRR) · Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Research Policy, 42(9), 1568-1580.

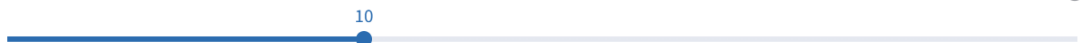
[執行新分析](#) 資料與屬性 編碼複核 交互分析 主題結構 詞庫誘導 信度檢定 匯出 框架建構

段落式或表格式都支援。表格若含「講者／內容」欄位會自動辨識。講者標記對信度檢定模組很重要。

上傳逐字稿 (.docx)



每份稿件間隔秒數



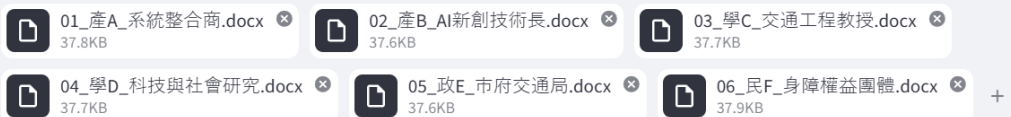
> 編碼框架: Responsible Innovation (AIRR)

開始分析

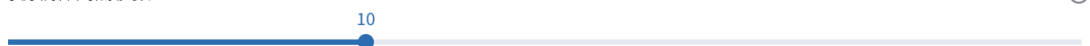
Figure 4. Upload one or more .docx transcripts. The delay slider sets the pause between files to avoid rate limits; 10 seconds is sensible on a free tier.

段落式或表格式都支援。表格若含「講者／內容」欄位會自動辨識。講者標記對信度檢定模組很重要。

上傳逐字稿 (.docx)



每份稿件間隔秒數



> 編碼框架: Responsible Innovation (AIRR)

開始分析

01_產A_系統整合商.docx ⇄

AI 編碼中... (第 1/3 次嘗試)

Figure 5. Each transcript is retried up to three times on malformed output or network failure.

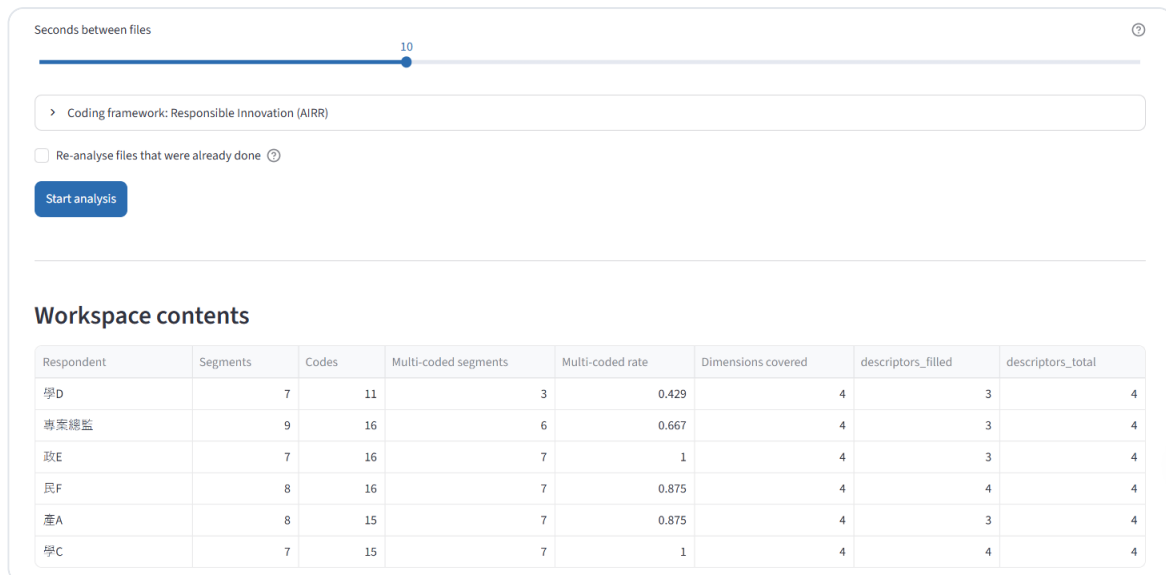


Figure 6. Six transcripts completed. Each reports the number of coded segments and how many carry more than one code.

Why multi-coding matters

A single passage evidencing two dimensions at once is normal in qualitative data, not noise. Forcing one code per segment destroys co-occurrence analysis — and co-occurrence is precisely where findings like "these two dimensions are structurally bound together in respondents' accounts" come from. The data structure here is flat: one segment may carry several codes.

If quota runs out mid-run, the batch stops and reports how many finished and how many never started. Swap in another key or wait for the quota to reset, keep the same file selection, and press Start again: completed transcripts are skipped automatically and are never billed twice.

2. Loading saved analyses

載入既有資料

只載入這一次要納入分析的訪談稿。其他頁籤的所有分析都以此處載入的內容為範圍，選得少一點，比較起來會清楚許多。

全選全不選重新掃描

要載入的訪談稿

學C (7) · 學C_20260... ×

學D (7) · 學D_20260... ×

政E (7) · 政E_20260... ×

民F (8) × 產A (8) ×

專案總監 (8) ×

×

▼

載入所選

已載入 6 / 25

清空工作區

目前分析中的受訪者

6

Figure 7. The picker lists each saved analysis with its segment count. Where a respondent has several saved runs, the filename is appended so the entries can be told apart.

Why selective loading

Every analysis downstream is scoped to what is currently loaded. Loading twenty interviews at once makes crosstabs and co-occurrence matrices unreadable, and usually you want to compare three or four. Scanning is not loading: the workspace starts empty and fills only when you press Load.

3. Data and descriptors

Respondent descriptors

Filled in automatically from the transcript; anything not stated is left Unspecified. Edit below and save. Descriptors are the grouping variable for cross-analysis, so review them before running any cr

Respondent	Institution type	Role level	Sector	Experience	Basis (from AI)
學D	Academia	Researcher	Other	Unspecified	訪談逐字稿：學D（科技與社會研究 博士後研究員）
專案總監	Industry	Middle management	ICT / AI	Unspecified	訪談逐字稿：產A（系統整合商 專案總監）
政E	Government	Middle management	Other	Unspecified	訪談逐字稿：政E（市政府交通局 科長）
民F	Non-profit	Senior management	Other	Over 20 years	訪談逐字稿：民F（身障權益倡議團體 秘書長），我做這個工作二十幾年，
產A	Industry	Middle management	ICT / AI	Unspecified	訪談逐字稿：產A（系統整合商 專案總監）
學C	Academia	Researcher	Other	Over 20 years	訪談逐字稿：學C（大學 交通工程教授），我做交通工程做了二十幾年

Save descriptors

Industry
Academia
Government
Non-profit
Research institute
Other

Data health

Respondent	Unspecified	des	Multi-coded segments	Multi-coded rate	Dimensions covered	descriptors_filled	descriptors_total	
學D		7	11	3	0.429	4	3	4
專案總監		9	16	6	0.667	4	3	4
政E		7	16	7	1	4	3	4

Figure 8. Descriptors drive every later cross-analysis. The model fills them only from information the transcript **states explicitly**; anything not stated is left Unspecified. It never infers a descriptor from tone or topic, because inferred descriptors silently contaminate group comparisons.

資料健檢

受訪者	段落數	編碼數	多重編碼段落	多重編碼比例	涵蓋維度數	descriptors_filled	descriptors_total
學C	7	12	4	0.571	4	4	4
學D	7	11	4	0.571	4	3	4
政E	7	16	7	1	4	3	4
民F	8	17	7	0.875	4	4	4
產A	8	18	8	1	4	3	4
專案總監	8	15	6	0.75	4	3	4

Figure 9. Data health. These figures tell you whether a transcript can support downstream analysis at all: too few segments or too few dimensions covered means there is little to say under this framework.

4. Code review: interpretive authority

Methodologically this is the most important tab. If the paper is to claim that all coding was human-verified, the review rate must reach 100% here.

▼ 為什麼需要這一頁

「質性詮釋不該外包給機器」是認識論上的反對，統計量回應不了，只能用流程回應：模型做初編碼，研究者確認、修正或刪除每一個編碼，並補入模型漏標的段落。本頁的每個數字都是這句話的證據。

複核者姓名（寫入稽核軌跡）

Hung-Chi Chang 張弘佑

[逐段複核](#) [人工新增](#) [引文逐字驗證](#) [複核報表](#)

受訪者

專案總監

☒ 只顯示未審核

篩選編碼

全部

專案總監 已複核 0/8

第幾頁（每頁 10 筆） / 2

1

S001 未審核 AI 原始: ANT-N/REF-N

標題

社會影響非守備範圍

精簡引文

可是你說社會影響喔.....老實講，那不是我們的守備範圍。我們是承包商，市府給規格，我們把它做出來，就這樣。

完整原文

Figure 10. Enter a **reviewer name** first — every change is recorded against it. Approval must be attributable.

S001 未審核 AI 原始: ANT-N/REF-N

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完整原文

嗯.....其實我覺得這個問題要分兩個層次來看。技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我們內部都有一套驗收標準，做得比市府要求的還嚴。可是你說社會影響喔.....老實講，那不是我們的守備範圍。我們是承包商，市府給規格，我們把它做出來，就這樣。

編碼

ANT-N 預期·短視／決定論 (-) × REF-N 反思性·自負／過度自信 (-) ×

AI 給的理由: 明確表示不負責社會影響，將預測範圍限於技術規格。; 將角色定義為純技術承包商，缺乏對社會影響的自我審視。

確認

刪除

「確認」會一併套用你在上面改過的文字與編碼；沒有改動時記為確認未改。

Figure 11. Expanding a segment shows the model's codes and its stated rationale. You may confirm, re-code, or delete the segment, and you may add segments the model missed.

Why the model's original codes are retained

After a change, the model's original judgement remains in the record. This lets you answer a question reviewers reliably ask — how wrong was the model? The modification rate is itself a reportable figure. A rate around forty percent usually indicates the framework definitions need sharpening rather than that the model is unusable.

變動段落前後對照

受訪者	段落ID	狀態	ai_original_codes	codes_after_review	removed	added	精簡引文	edit_c
專案總監	M002	added		ANT-P		ANT-P	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我	
專案總監	M001	deleted		ANT-P		ANT-P	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我	

稽核軌跡

time	受訪者	段落ID	action	detail	reviewer	狀態	精簡引文
2026-08-15T11:58:16	專案總監	M001	add	model miss	Hung-Chi Chang	deleted	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我
2026-08-15T12:00:43	專案總監	M002	add	model miss	Hung-Chi Chang	added	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我
2026-08-15T12:01:36	專案總監	M001	delete		Hung-Chi Chang	deleted	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我
2026-08-15T12:06:47	專案總監	S001	update_t	quote:"可是	Hung-Chi Chang	confirme	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我
2026-08-15T12:06:47	專案總監	S001	confirm		Hung-Chi Chang	confirme	技術層次上，我們一定會做完整的測試，這個沒有問題，那個誤判率、辨識率，我

下載－複核報表

Figure 12. The audit trail: who changed what, when, and from what to what. Exportable as an appendix.



Figure 13. Review statistics, with a ready-made methods paragraph in both languages. Every number comes from the recorded trail, not from an estimate.

5. Cross-analysis

5.1 Dimension × descriptor crosstabs



Figure 14. Row percentages are shown by default rather than raw counts: groups differ in size, and raw counts cannot be compared across them.

列百分比 = 各屬性組別內部的編碼分布，已消除組別大小差異，組別之間才可直接比較。原始次數受該組人數與逐字稿長度影響。

▼ 卡方檢定 (謹慎解讀)

χ^2	df	Cramér's V	N
40.136	21	0.386	90

28/32 格期望次數 < 5 (超過 20% 上限)，卡方前提不成立，因此不提供 p 值，請僅以描述性方式解讀交叉表。

另請注意：卡方假設觀察值彼此獨立，但同一位受訪者貢獻多筆編碼會違反此假設。質性樣本下建議把統計量當作型態描述，而非顯著性推論。

Figure 15. Here the tool has determined that expected cell counts are too low and **refuses to report a p-value**, offering descriptive interpretation only.

Why withholding the p-value is the right behaviour

Qualitative samples typically run from a handful to a few dozen respondents, so most cells fall below an expected count of 5 and the chi-square assumptions simply do not hold. Printing a p-value anyway invites a claim that cannot survive scrutiny. If a reviewer asks how a chi-square is justified at this n, there is no good answer — so the tool declines to produce one.

5.2 Code co-occurrence

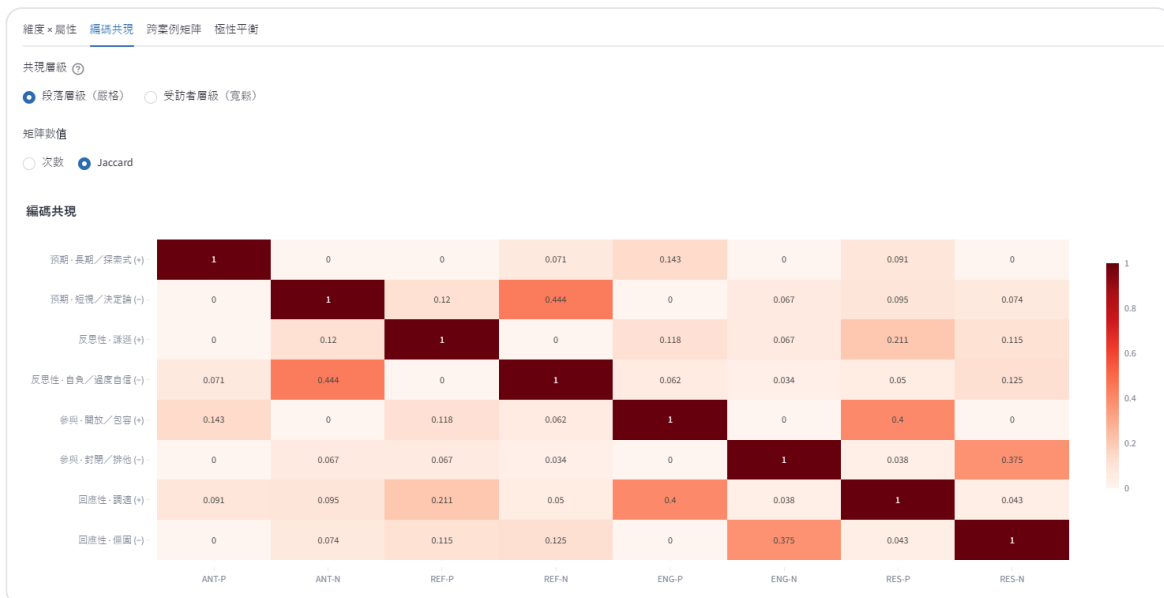


Figure 16. Switch between **segment level** (strict: both codes in the same passage) and **case level** (loose: the same respondent said both at some point).

主要共現配對

code_a	code_b	cooccurrence	count_a	count_b	Jaccard	cross_dimension
參與・封閉／排他 (-)	回應性・偏固 (-)	9	18	15	0.375	☑
預期・短視／決定論 (-)	反思性・自負／過度自信 (-)	8	14	12	0.444	☑
參與・開放／包容 (+)	回應性・調適 (+)	4	5	9	0.4	☑
反思性・謙遜 (+)	回應性・調適 (+)	4	14	9	0.211	☑
反思性・自負／過度自信 (-)	回應性・偏固 (-)	3	12	15	0.125	☑
預期・短視／決定論 (-)	反思性・謙遜 (+)	3	14	14	0.12	☑
反思性・謙遜 (+)	回應性・偏固 (-)	3	14	15	0.115	☑
反思性・謙遜 (+)	參與・開放／包容 (+)	2	14	5	0.118	☑
預期・短視／決定論 (-)	回應性・調適 (+)	2	14	9	0.095	☑
預期・短視／決定論 (-)	回應性・偏固 (-)	2	14	15	0.074	☑

Jaccard = 共現次數 ÷ 兩碼聯集次數。可校正「某碼本來就常常出現」造成的假性強關聯。跨維度的配對最具理論意義，代表兩個維度在同一段陳述中結構性地綁在一起。

多重編碼段落：36 / 46

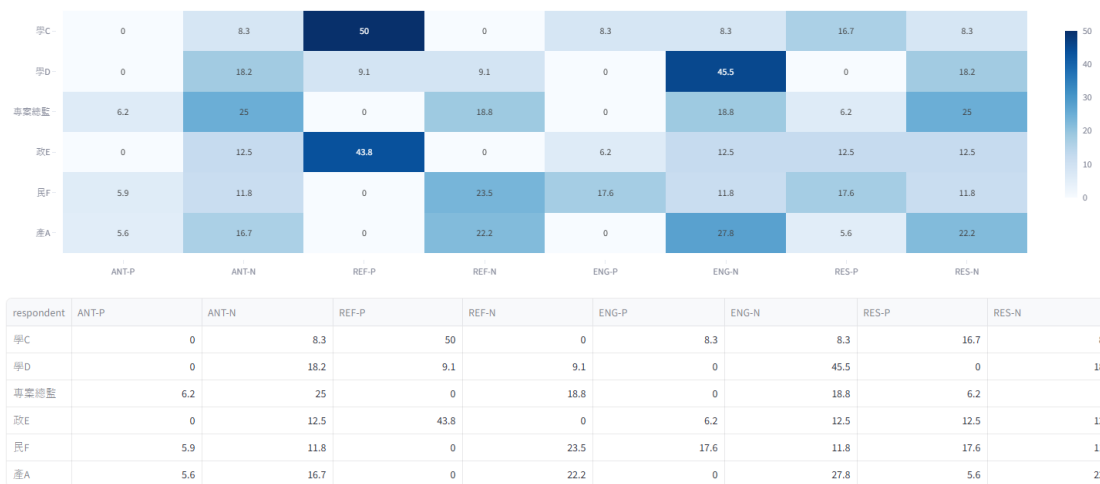
Figure 17. Pairs flagged `cross_dimension` carry the most theoretical weight: two **different** dimensions bound together within a single utterance.

Why Jaccard rather than raw counts. Jaccard is co-occurrence divided by the union of the two codes. A code that is simply frequent will co-occur often with everything, which is a spurious association; Jaccard removes that effect.

5.3 Cross-case matrix

☒ 以受訪者為單位標準化為百分比 ①

跨案例矩陣



列 = 受訪者，欄 = 編碼。橫向看是個案的編碼輪廓，縱向看是該編碼在樣本中的分布。整欄接近零代表該理論面向在本樣本中缺乏證據，這本身就是值得討論的發現。

Figure 18. Respondent × code. Normalised, each cell is the share of that respondent's codes, which removes differences in how much each person talked.

5.4 Polarity balance

Polarity is an extension specific to the responsible innovation framework: a dimension can be **enacted** (+) or **undermined** (–). Frameworks without polarity do not show this sub-tab at all.

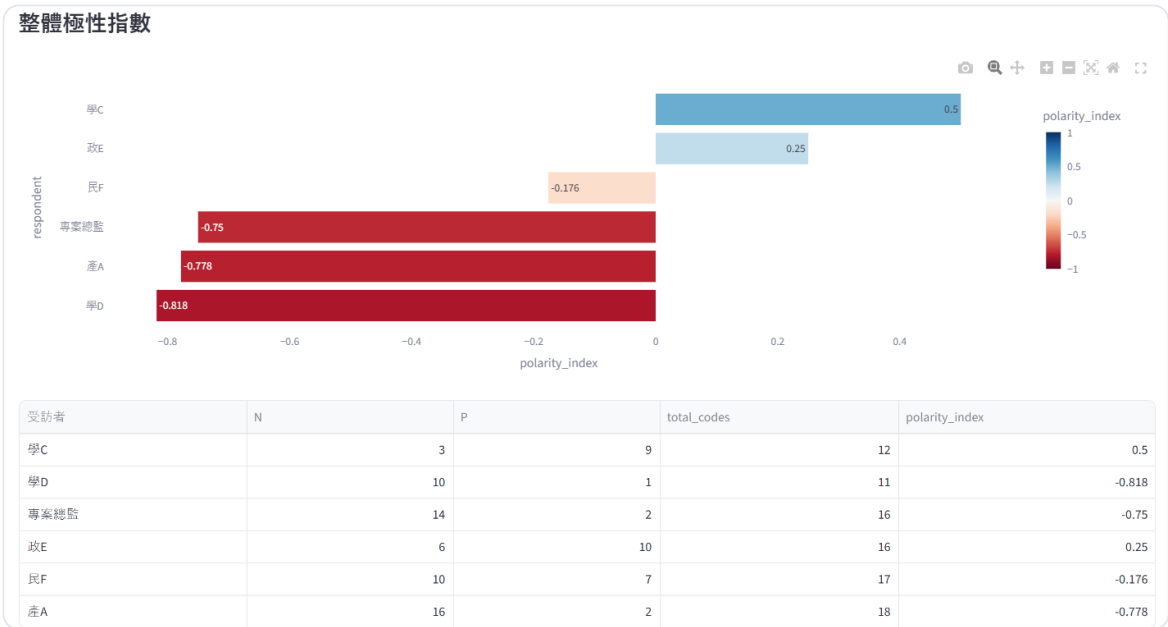


Figure 19. Polarity index = $(P-N)/(P+N)$, bounded –1 to +1 and normalised for coding volume, so cases are comparable.

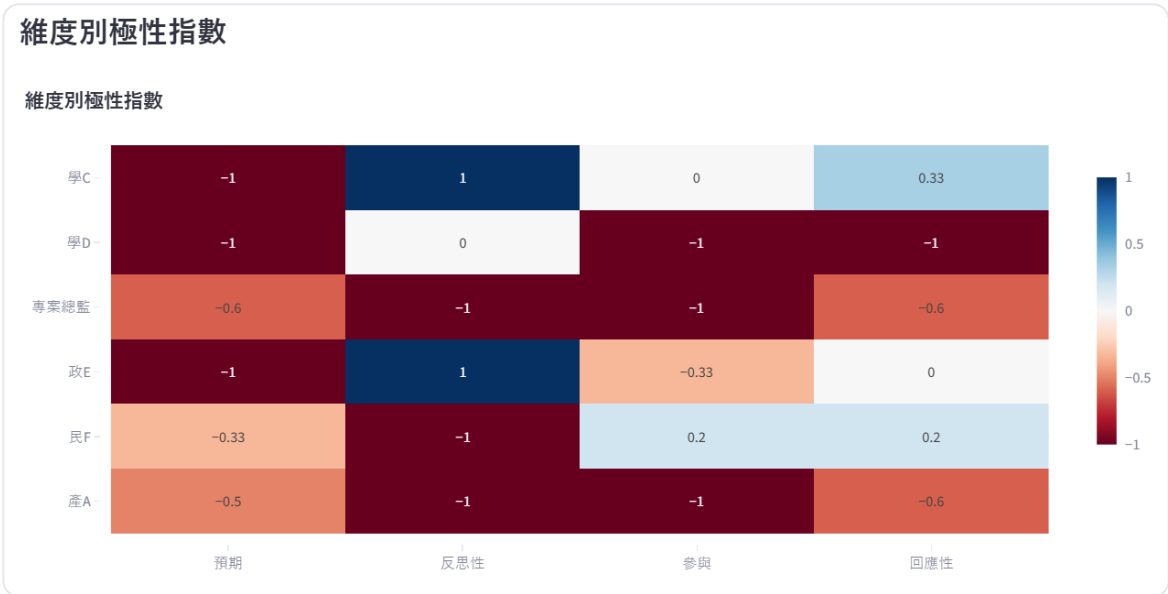


Figure 20. Per-dimension heatmap: where a respondent is markedly negative becomes visible at a glance.

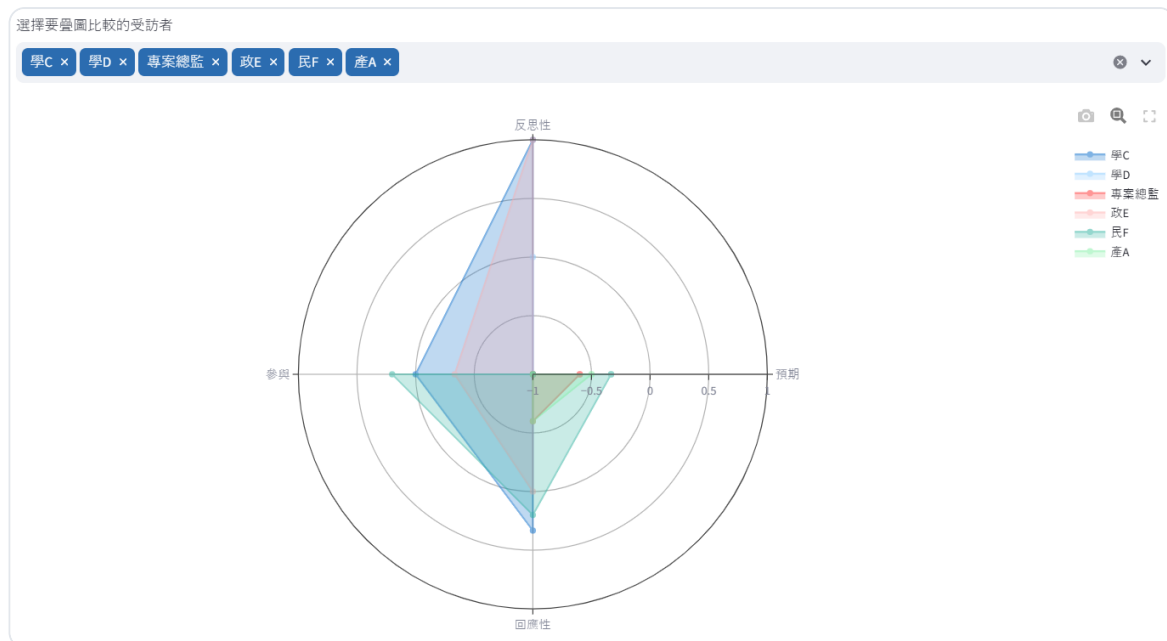


Figure 21. Radar overlay, useful for contrasting group profiles.

受訪者	維度	N	P	total	pn_ratio	polarity_index
學C	預期	1	0	1	0	-1
學C	參與	1	1	2	1	0
學C	反思性	0	6	6	None	1
學C	回應性	1	2	3	2	0.333
學D	預期	2	0	2	0	-1
學D	參與	5	0	5	0	-1
學D	反思性	1	1	2	1	0
學D	回應性	2	0	2	0	-1
專案總監	預期	4	1	5	0.25	-0.6
專案總監	參與	3	0	3	0	-1

極性指數 = (P-N)/(P+N)。已對編碼總量標準化，可跨案例比較；但當某維度的編碼總數過少（如 <3），指數會非常不穩定，解讀時請對照「總數」欄。

Figure 22. The long table. **Always read the index against the total column** — with only one or two codes in a dimension the index is extremely unstable.

A polarity index of -1 does not mean a respondent is bad. It means every coded statement of theirs in that dimension fell on the undermining side. It is a property of the text, not an evaluation of a person, and the distinction belongs in the write-up.

6. Theme structure: Gioia-style data structure

Everything so far has been deductive — reading data through an existing framework. This tab inverts that: themes are induced from the data, then read back against the framework.

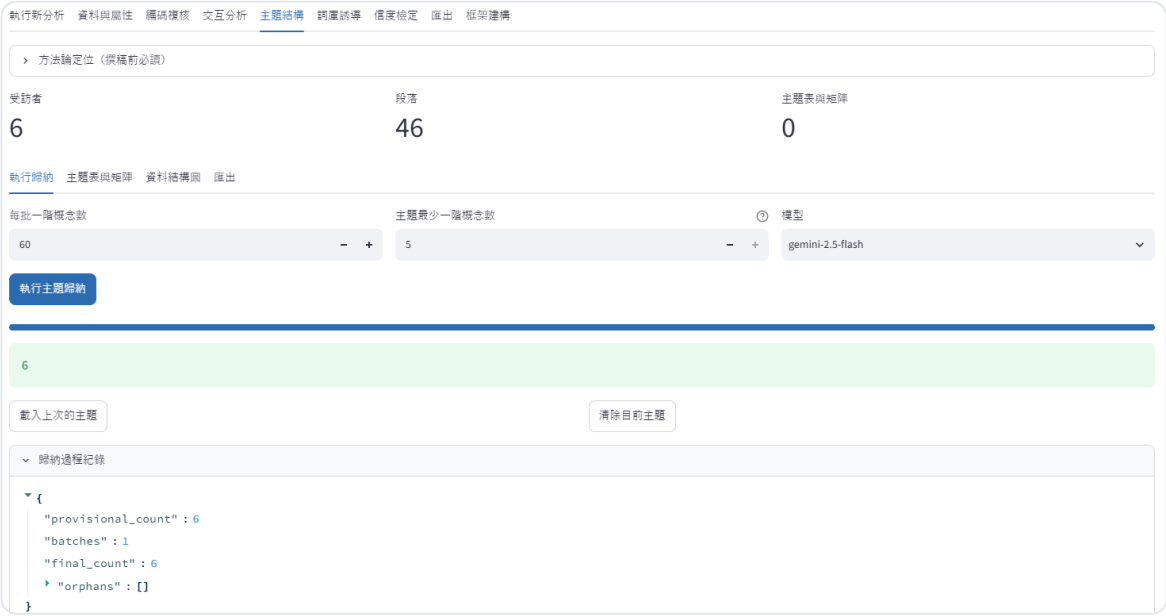


Figure 23. Two stages: first-order concepts are batched into provisional themes, then provisional themes are consolidated. The process log keeps every intermediate result for auditing.

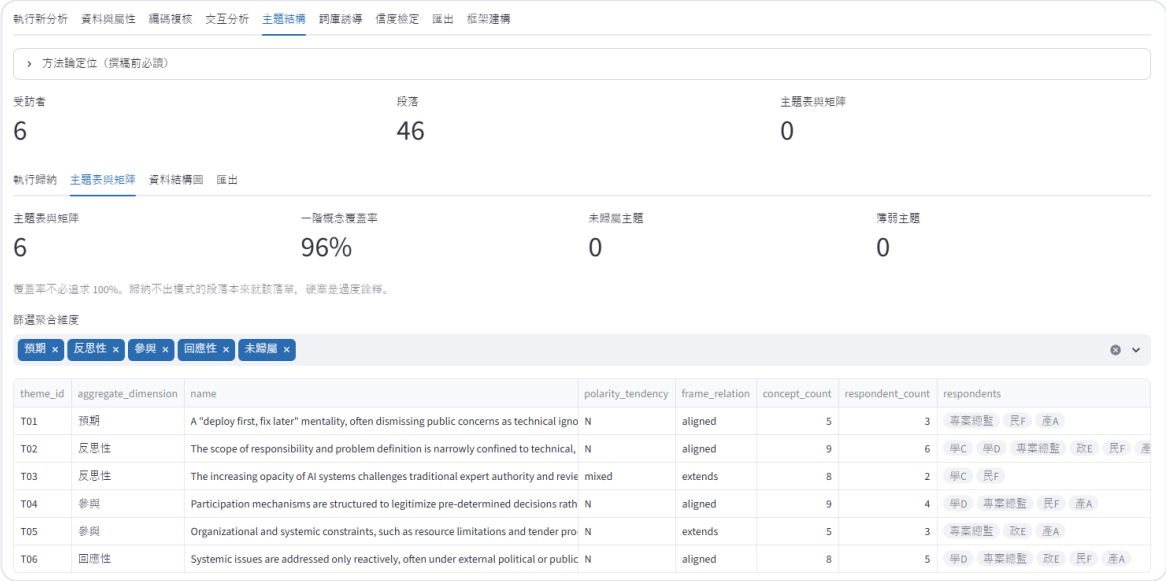


Figure 24. The `frame_relation` column is the one to read: `aligned` means the theme sits inside what the framework anticipates, `extends` means it exceeds the framework — which is where new contributions live.

Coverage need not approach 100%. Segments that form no pattern should remain unassigned; forcing them into a theme is over-interpretation. Unassigned first-order concepts are listed for inspection.

歸納主題 × 演繹編碼

theme_id	aggregate_dimension	name	REF-N	RES-N	RES-P	ANT-N	ENG-N	ANT-P	REF-P	ENC
T01	預期	A "deploy first, fix later" mentality, often dismissing public concerns as technical ignorance, prioritizes rapid deployment and technical solutions over comprehensive foresight and proactive problem-solving.	4	2	1	2	1	1	0	
T02	反思性	The scope of responsibility and problem definition is narrowly confined to technical, engineering, or operational concerns, often overlooking broader societal or ethical implications.	6	1	0	7	0	1	3	
T03	反思性	The increasing opacity of AI systems challenges traditional expert authority and review mechanisms, necessitating more transparent and accountable processes.	2	1	2	2	1	0	5	
T04	參與	Participation mechanisms are structured to legitimize pre-determined decisions rather than to genuinely incorporate diverse stakeholder perspectives.	0	3	1	1	8	1	0	
T05	參與	Organizational and systemic constraints, such as resource limitations and tender processes, often hinder the implementation of more inclusive participation models.	0	3	1	0	3	0	3	
T06	回應性	Systemic issues are addressed only reactively, often under external political or public pressure, rather than through proactive systemic reform.	0	5	3	2	5	0	2	

橫跨多個編碼的主題最值得注意——代表它切過了框架的分類邊界。

未納入任何主題的一階概念 (2)

global_id	受訪者	標題	精簡引文
G0007	學D	研究立場與自我反思	我做這個領域，我的預設就是會去找權力不對等的地方，所以我看到的東西跟工程師看到的不一樣。
G0029	民F	堅持參與以爭取微小進展	說老實話，我的期待不高，但是我還是會繼續去開會。因為我如果不去，連那一行「部分民眾」都看不見。

Figure 25. Induced themes × deductive codes. A theme spanning several codes is the interesting case: it cuts across the framework's own boundaries.

The data structure figure

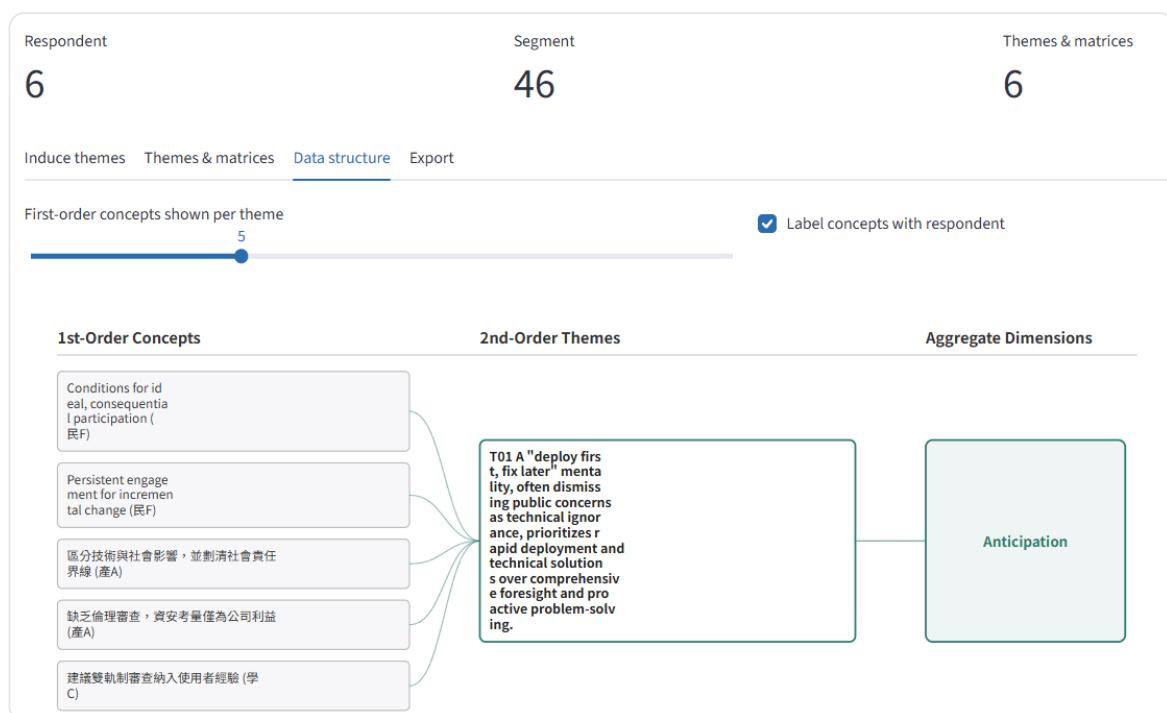


Figure 26. First-order concepts, second-order themes, aggregate dimensions — the standard Gioia presentation.

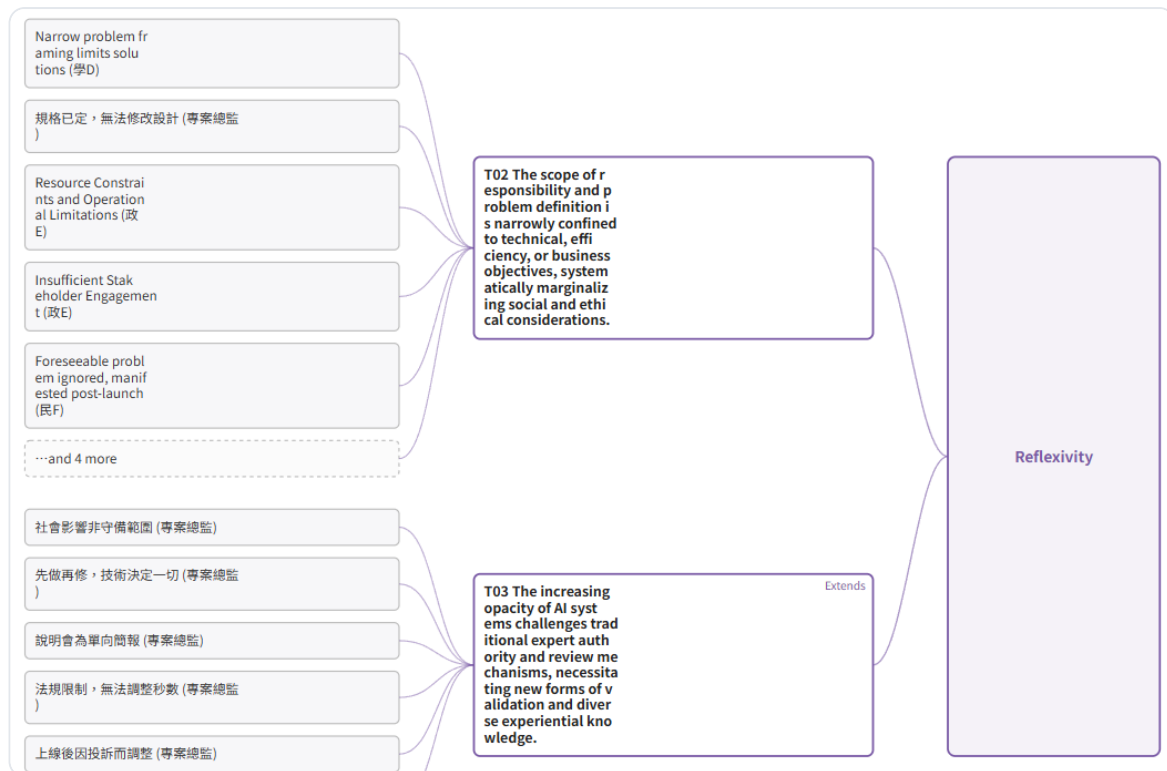


Figure 27. Themes marked as extending the framework are the richest material for a discussion section.

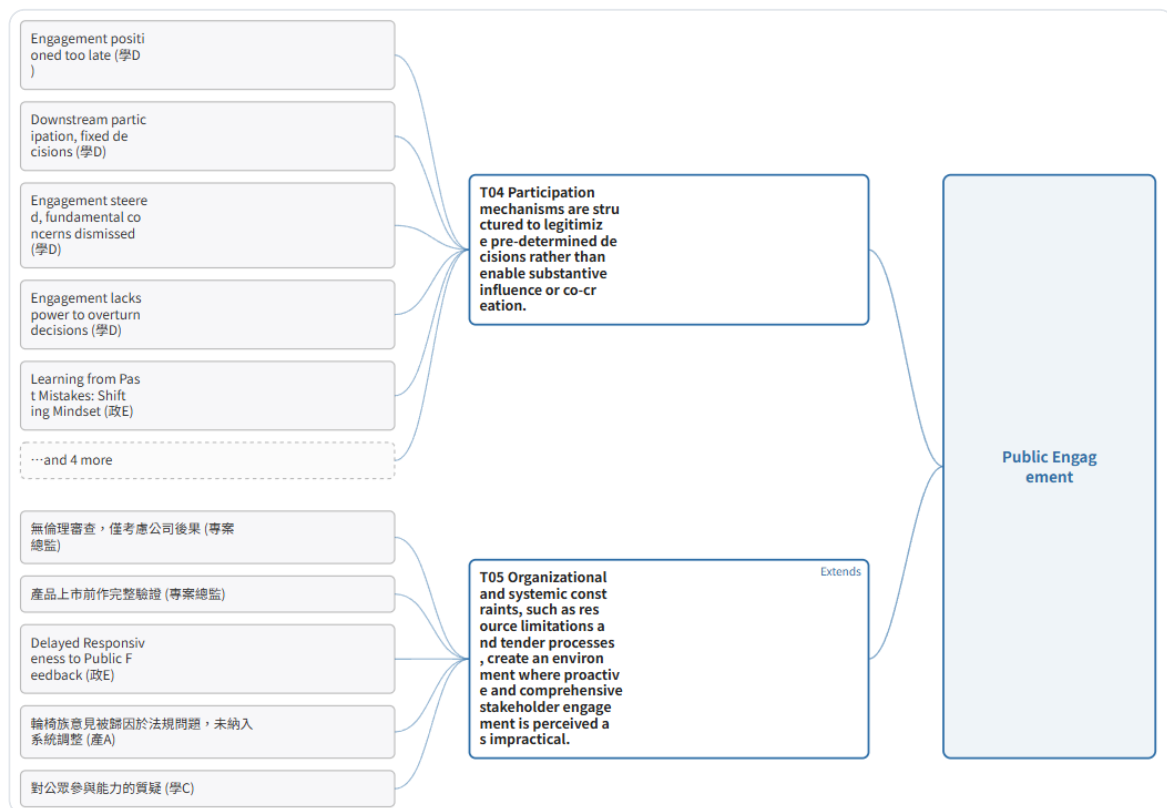


Figure 28. Two themes under the engagement dimension.

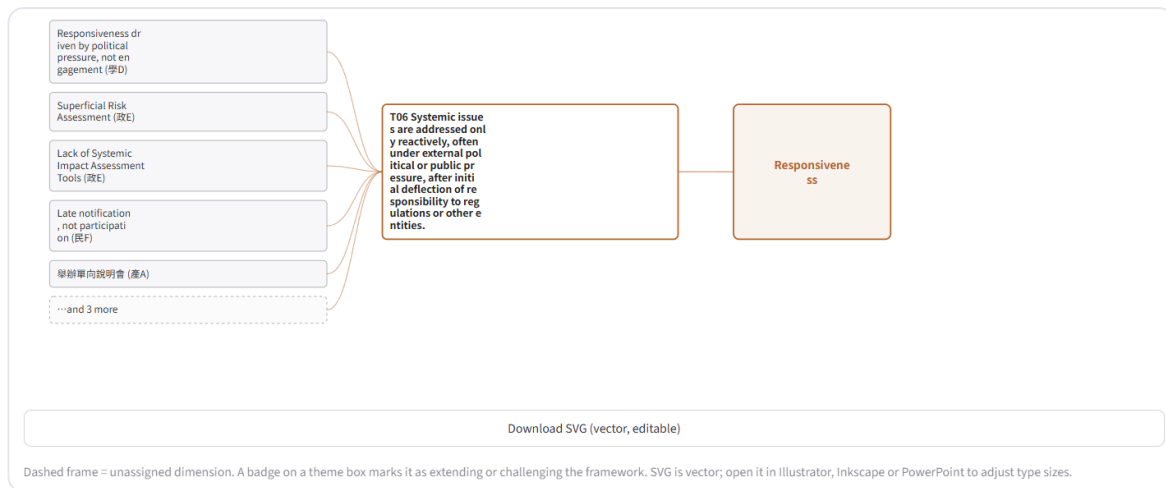


Figure 29. The final section, with SVG export. SVG is vector, so it can be opened in Illustrator, Inkscape or PowerPoint to adjust type sizes before publication.

Theme names appear in English in these screenshots because the version used did not pass the analysis output language into the second induction stage. **This has been fixed:** Chinese material now yields Chinese theme names, and the live figure is cleaner than shown.

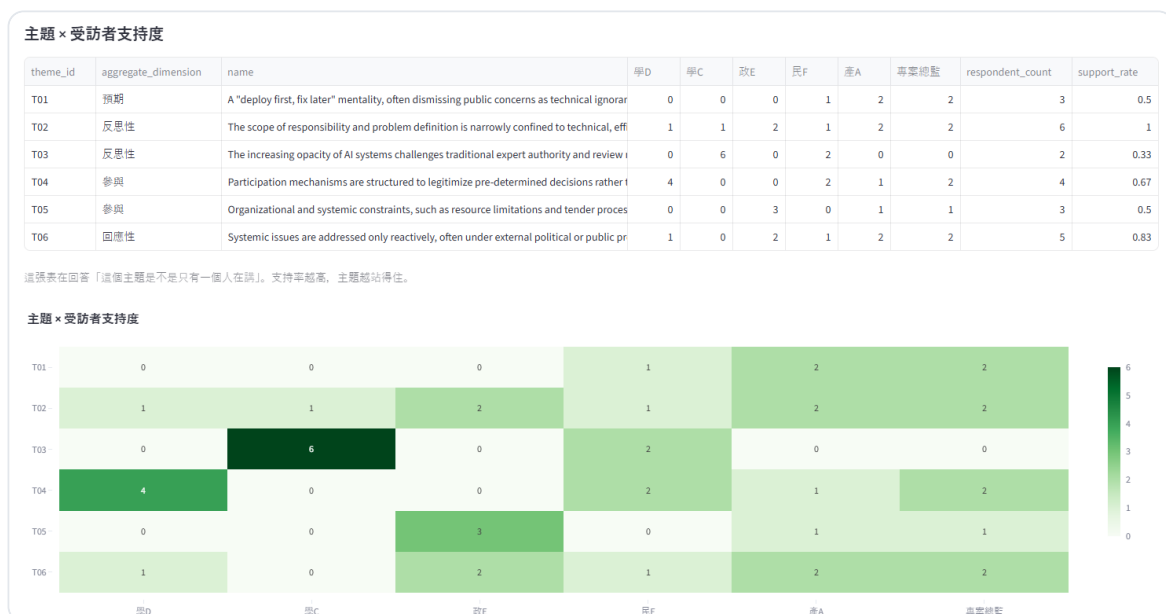


Figure 30. Theme × respondent support. Treat a low `support_rate` with suspicion — something only one respondent said is rarely a theme.

7. Lexicon induction

The stance here is explicit: **a lexicon is not a classifier**. Keyword matching as a primary coding mechanism is a return to bag-of-words and performs poorly on

Chinese. The lexicon is used for three things it is actually good at: discovering domain terms, inducing distinctive terms, and serving as a transparent baseline against which to audit the model.

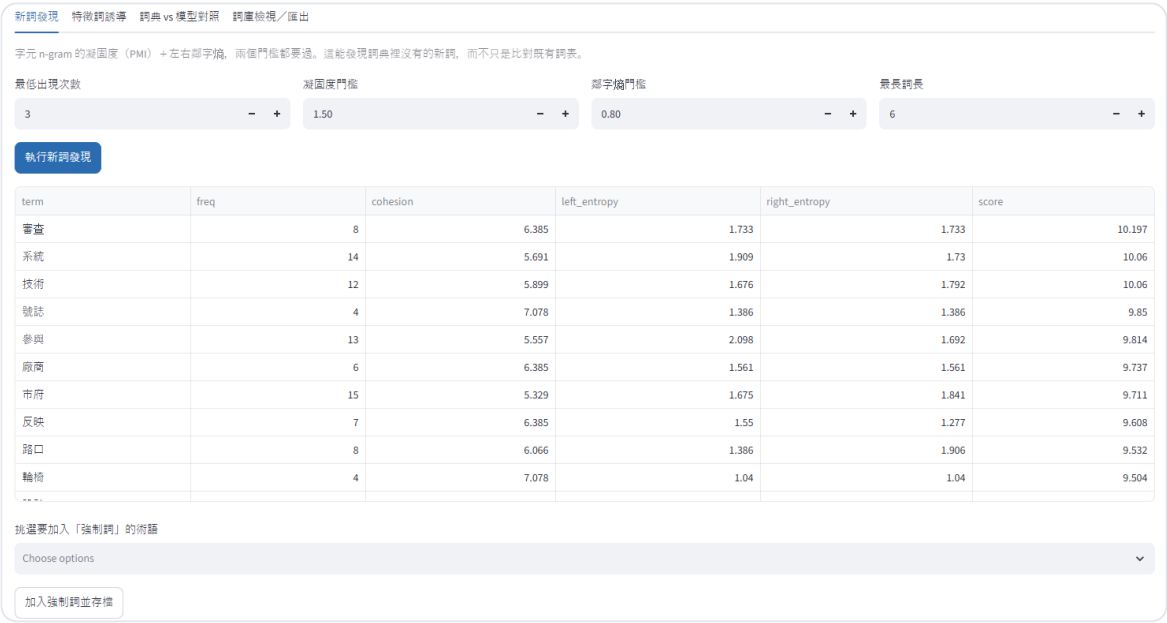


Figure 31. Term discovery runs without any segmenter: pointwise mutual information filters loose combinations, and left/right neighbour entropy filters incomplete fragments.



Figure 32. Feature induction uses log-odds with an informative Dirichlet prior (Monroe et al., 2008), which suppresses terms common to all codes and is not misled by rare ones.

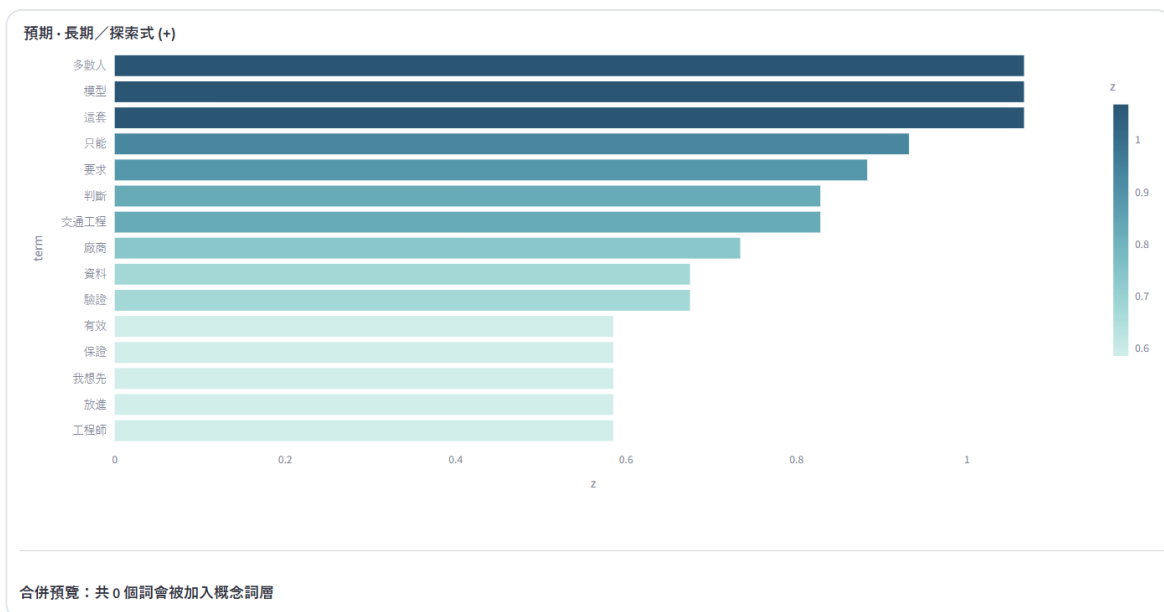


Figure 33. Here every z falls below the merge threshold of 2.00, so the merge preview contains no terms at all.

The demonstration corpus is too small for significant feature terms, which is itself instructive: terms with z below 2 do not belong in a lexicon. On a real corpus genuine domain vocabulary surfaces here.

Dictionary versus model



Figure 34. This screenshot shows a **fault condition**: overall $\kappa = 0.0$ and every `dict_count` is zero. The dictionary coder produced nothing at all.

This figure is kept because it documents a real defect.

The application looked for a lexicon file named `lexicon_ri_stilgoe_2013_zh-Hant.json` while the actual file was `ri_lexicon_zh.json`. Finding nothing, it created an empty lexicon and subsequently wrote that empty lexicon back to disk, overwriting 239 concept terms and 17 syntactic patterns.

Both problems are fixed: the filename mapping is corrected, and overwriting a non-empty lexicon with an empty one is now refused outright. Working installations show real numbers here.

In normal operation this page is diagnostic: codes with low κ are usually exactly where dictionary methods fail on Chinese — negation, irony, reported speech. Those concrete cases are the evidence for arguing that keyword methods are insufficient.

8. Inter-rater reliability

RI-AutoCoder PRO

Responsible Innovation (AIRR) · Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. *Research Policy*, 42(9), 1568-1580.

Run analysis · Data & descriptors · Code review · Cross-analysis · Theme structure · Lexicon induction · Reliability · Export · Framework builder

> Trap 1 — the sampling frame must include uncoded units

> Trap 2 — do not report Cohen's kappa alone

Build sampling frame · Sample & blind · Collect codings · Reliability report

Transcripts for all 6 respondents are already stored with their analyses. Nothing to upload.

Minimum unit length: 15 — + Maximum unit length: 400 — +

Build sampling frame

Figure 35. The two collapsed panels at the top document the methodological traps this page is built to avoid.

If the second coder only sees passages the model already coded, recall is identically 1, because the frame contains nothing the model could have missed. Once unmarked units are included, anything the human coded and the model did not appears as a false negative. **That FN count is the "how much did the model miss" figure reviewers want.**

stratum	n
未標記	35
ENG-N+RES-N	10
ENG-N	6
ANT-N+REF-N	4
REF-P	4
ANT-N+REF-P	3
REF-P+RES-N	2
ANT-N+REF-N+RES-N	2
ENG-P+REF-P+RES-P	2
REF-P+RES-P	2

編碼表刻意不含模型編碼與分層資訊，確保雙盲。該碼成立就填 1 (v、√、x 也認得)，不成立留空。刻意設計成離線填寫——合作者不該為了編碼去安裝任何東西。

下載 — coder_b

Figure 36. Stratification, and one downloadable coding sheet per coder.

unit_id	A	B	C	D	E	F	G	H	I	J	K	L
1	U0020	coders	會，我在期中審查的意見書裡面寫過。可是我後	ANT-P	ANT-N	REF-P	REF-N	ENG-P	ENG-N	RES-P	RES-N	comment
2	U0088	coders	倫理審查是沒有。但是我們有做資安風險評估	v	v	v				v		
3	U0034	coders	最明顯的就是溝通。理想上我們應該在規劃階段			v			v			
4	U0047	coders	第一件他們說是法規規定，叫我去跟中央講。第								v	
5	U0017	coders	您擔任這個案子的技術顧問，可以談談您的角色									
6	U0065	coders	有一位輪椅族的代表，講說行人穿越的秒數對他								v	
7	U0061	coders	那在系統設計的過程中，有邀請民間團體或一般									
8	U0030	coders	案例：濱海市智慧號誌與行人偵測系統 ※									
9	U0059	coders	在專案初期，有沒有討論過這個系統可能產生哪									
10	U0081	coders	那時候規格已經確定了，因為要發包囉，招標文						v		v	
11	U0049	coders	我做這個工作二十幾年，我的看法是，參與最重					v		v		
12	U0010	coders	我不會說它是假的，主辦的人是真心想做好。問			v			v			
13	U0011	coders	那您怎麼看市府後來加裝延長按鈕的回應？									
14	U0083	coders	後來有針對這個意見做任何調整嗎？									
15	U0046	coders	有，我講了兩件事。第一件是行人穿越的秒數，	v				v				
16	U0021	coders	您怎麼看這個系統對不同族群的影響？									
17	U0003	coders	我做的是科技與社會研究，這個案子我是從外部						v			
18	U0016	coders	案例：濱海市智慧號誌與行人偵測系統 ※									
19	U0084	coders	後來……嗯，其實上線之後我們有加一個功能，			v				v		
20	U0053	coders	我想了很多年。我覺得根源是，設計這些系統的			v		v				
21	U0007	coders	您參加的公民咖啡館，運作情況如何？									
22	U0031	coders	想請您談談市府在這個專案裡面的角色。									
23	U0080	coders	說明會是在設計定案之前還是之後？									
24	U0002	coders	案例：濱海市智慧號誌與行人偵測系統 ※									
25	U0037	coders	因為沒有範本。我們寫公文是有範本的，環評有	v		v						
26	U0018	coders	我是評選委員，後來也在期中審查提供意見。不			v						

Figure 37. The downloaded workbook contains quotations and blank code columns — **the model's answers are absent**. This is what makes the procedure genuinely blind.

The header of column C renders as repeated `texttexttext...` in this capture; the correct field name is `text`.

建立抽樣框 抽樣與盲測 回收編碼 信度報表

coder_a — 已回收 83/83

上傳 coder_a 填好的編碼表

irr_7f...oder_a.xlsx 22.8KB +

匯入 coder_a

coder_b — 已回收 83/83

上傳 coder_b 填好的編碼表

irr_7f...oder_b.xlsx 20.4KB +

匯入 coder_b

Figure 38. Completed sheets are uploaded back. Coding can also be done directly in the interface.

建立抽樣框 抽樣與盲測 回收編碼 信度報表

編碼者 A 編碼者 B

coder_a coder_b

共同完成 83 個單元

觀察一致率	Cohen's κ	PABAK	Gwet's AC1
0.935	0.675	0.87	0.919

PABAK = 0.87、AC1 = 0.919, 三者方向一致。

整組編碼完全相同的比例

0.663

Figure 39. Four coefficients side by side: observed agreement, Cohen's κ , PABAK and Gwet's AC1.

Why four coefficients rather than one

Cohen's κ is severely deflated when most units are judged absent by both coders — the kappa paradox — and qualitative coding meets this condition almost by construction, because marking is sparse. PABAK and AC1 are insensitive to prevalence, so reporting all three lets a reader judge whether a low κ reflects coder disagreement or sparse marking.

盛行率偏高（多數單元雙方都判定為「無」），屬典型的 kappa paradox：PABAK = 0.777、AC1 = 0.857 明顯較高。報告時應三個係數並陳，並說明 κ 偏低來自標記稀疏而非編碼者不一致。

Figure 40. When the paradox is detected the tool raises it explicitly and states how to report it.

維度混淆矩陣

	coder_b: 預期	coder_b: 反思性	coder_b: 參與	coder_b: 回應性	coder_b: 無
coder_a: 預期	11	6	6	5	0
coder_a: 反思性	9	15	5	7	0
coder_a: 參與	5	5	18	7	0
coder_a: 回應性	5	3	8	11	0
coder_a: 無	2	2	2	4	38

極性一致度（雙方都判定屬該維度的單元）

維度	both_assigned	polarity_agree	polarity_conflict	polarity_agreement_rate
預期	11	8	3	0.727
反思性	15	15	0	1
參與	18	18	0	1
回應性	11	9	2	0.818

極性分歧比維度分歧嚴重——代表雙方從同一句話讀出了相反的評價方向，這是編碼手冊需要補充定義的地方。

Figure 41. Dimension and polarity confusion are shown separately. **Polarity disagreement is the more serious kind:** the two coders read opposite evaluative directions from the same sentence, which means the codebook needs a sharper definition.

極性分歧比維度分歧嚴重——代表雙方從同一句話讀出了相反的評價方向，這是編碼手冊需要補充定義的地方。

分歧清單（校準會議用）

unit_id	respondent	coder_a	coder_b	codes_a	codes_b	only_a	only_b	disagreement_type	text
U0037	政E	coder_a	coder_b	ANT-P	ANT-N REF-P	ANT-P	ANT-N	polarity_conflict	因為沒有範本。我們寫公文是有範本的
U0040	政E	coder_a	coder_b	REF-P	REF-P RES-N RES-P		RES-N	polarity_conflict	對。這件事情給我的教訓很深。第一次
U0071	產A	coder_a	coder_b	ANT-N	ANT-N REF-P	ANT-P		polarity_conflict	倫理審查是沒有。但是我們有做資安的
U0010	學D	coder_a	coder_b	ENG-N	ENG-N RES-N	REF-P	RES-N	dimension_mismatch	我不會說它是假的。主辦的人是真心想
U0046	民F	coder_a	coder_b	ANT-P	ENG-P	ANT-P		dimension_mismatch	有，我講了兩件事。第一件是行人穿越
U0084	專案總監	coder_a	coder_b	REF-P	ANT-N RES-P	REF-P	ANT-N	dimension_mismatch	後來……嗯，其實上線之後我們有加一
U0053	民F	coder_a	coder_b	ENG-P	ENG-P	REF-P		dimension_mismatch	我想了很多年。我覺得根源是，設計這
U0052	民F	coder_a	coder_b	ANT-P	ANT-N RES-P	ANT-P	ANT-N RES-P	dimension_mismatch	要分開講。市府後來加裝按鈕，這個和
U0077	專案總監	coder_a	coder_b	ANT-N	ANT-N REF-N	ENG-N		dimension_mismatch	非預期的後果。嗯，我這樣講你可能有
U0041	政E	coder_a	coder_b	ANT-P	REF-P RES-P	ANT-P		dimension_mismatch	我們現在在寫的新案子，我有把「利

Figure 42. The disagreement list doubles as the agenda for a calibration meeting.

Disagreements (agenda for the calibration meeting)

unit_id	respondent	coder_a	coder_b	codes_a	codes_b	only_a	only_b	disagreement_type	text
U0006	學D	coder_a	AI	ANT-P	ANT-N REF-N	ANT-P	ANT-N REF-N	polarity_conflict	應該是在市府決定編這筆預算之前。當時的問題設定是
U0010	學D	coder_a	AI	ENG-N	ENG-N RES-N	REF-P	RES-N	dimension_mismatch	我不會說它是假的，主辦的人是真心想做好。問題在於
U0067	產A	coder_a	AI	REF-P	ENG-N RES-N	REF-P	ENG-N RES-N	dimension_mismatch	那時候規格已經確定了，因為要發包囉，招標文件出去
U0039	政E	coder_a	AI	REF-P	ANT-N REF-P	RES-P	ANT-N	dimension_mismatch	有做，但是我要誠實講，做得很表面。我們有寫一份效
U0052	民F	coder_a	AI	ANT-P	ENG-P RES-P	ANT-P	ENG-P RES-P	dimension_mismatch	我做這個工作二十幾年，我的看法是，參與最重要的不
U0037	政E	coder_a	AI	ANT-P	ENG-N RES-N	ANT-P	ENG-N RES-N	dimension_mismatch	最明顯的就是溝通。理想上我們應該在規劃階段就開好
U0086	學C	coder_a	AI	ANT-N	REF-P RES-P	ANT-N	REF-P RES-P	dimension_mismatch	你這樣問，我要想一下。嗯，應該不等價。我剛剛
U0049	民F	coder_a	AI	ENG-P	ANT-N REF-N	ENG-P	ANT-N REF-N	dimension_mismatch	有，我講了兩件事。第一件是行人穿越的秒數，現在的
U0054	民F	coder_a	AI	ENG-P	ANT-N RES-N	ENG-P	ANT-N RES-N	dimension_mismatch	有。上線之後，我們會員反映在中山路那個路口，輪椅
U0082	學C	coder_a	AI	RES-N	ANT-P REF-P	RES-N	ANT-P REF-P	dimension_mismatch	我認為這是最人的盲點。我們交通工程的傳統，是把人

Download reliability report

Figure 43. Model precision and recall. **The FN column is the count of segments the model missed**, and it is meaningful only because the frame included unmarked units.

9. Framework construction

This tab removes the constraint of the built-in framework: a coding framework can be constructed from literature with full provenance.

RI-AutoCoder PRO

Responsible Innovation (AIRR) · Stigloe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Research Policy, 42(9), 1568-1580.

執行新分析 資料與屬性 編碼複核 交互分析 主題結構 詞庫誘導 信度檢定 匯出 框架建構

編碼框架是整份研究的效度根基，因此絕不取自模型的記憶。文獻是知識來源，模型只做摘要與草擬，決定權在研究者。每個維度都必須引用實際檢索到的文獻，且必須經具名研究者核可才會進入框架。

現用框架 1. 檢索文獻 2. 草擬與核可 匯入／匯出

已設定 API 全額：每日 100,000 credits，約可進行 10,000 次檢索。

理論概念

例如：responsible innovation

理論名稱常有歧義，先限定主題可避免混入不相干領域的文獻。非必要，但建議先做。

尋找對應主題

奠基文獻（依被引次數） 近年文獻 近年的年數界定

20 15 6

只保留有摘要的文獻

開始檢索

Figure 44. The four sub-tabs: active framework, literature retrieval, drafting and approval, import/export.

Step 1 — retrieve

開始檢索

已檢索到 30 篇文獻 (其中 25 篇有摘要)

檢索到的文獻中有 96% 在標題或摘要提到查詢詞。

完整片語只找到 10 篇，因此自動放寬為「所有詞都要出現」而非完整片語。此調整已記入檢索紀錄。

檢索結果會快取在本機。快取鍵不含你的金鑰，因此快取資料夾可隨專案一起流通：其他研究者不需金鑰、不需連網，即可重現這次檢索。

#	年份	文獻	被引次數	摘要
W1	2013	Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Res	3046	✓
W2	2012	Owen, R., Macnaghten, P., & Stilgoe, J. (2012). Responsible research and innovation: From science in so	1591	✓
W3	2018	Rose, D. C., & Chilvers, J. (2018). Agriculture 4.0: Broadening Responsible Innovation in an Era of Smart	637	✓
W4	2016	Burget, M., Bardone, E., & Pedaste, M. (2016). Definitions and Conceptual Dimensions of Responsible Ri	487	
W5	2021	Rijswijk, K., et al. (2021). Digital transformation of agriculture and rural areas: A socio-cyber-physical sy	351	✓
W6	2013	Caverly, R. W. (2013). Responsible Innovation: Managing the Responsible Emergence of Science and Inr	273	✓
W7	2020	Muiderman, K., et al. (2020). Four approaches to anticipatory climate governance: Different conception	209	✓
W8	2019	Hadj, T. B. (2019). Effects of corporate social responsibility towards stakeholders and environmental m	204	
W9	2014	Nordmann, A. (2014). Responsible innovation, the art and craft of anticipation. Journal of Responsible	180	✓
W10	2010	Daneshvar, D. H., et al. (2010). Helmets and Mouth Guards: The Role of Personal Equipment in Preventi	175	

Figure 45. Retrieved works are numbered W1, W2 ... and those identifiers are what the hallucination guard operates on. Here the exact phrase matched too few works, so the query was widened to require all terms — a change recorded in the retrieval log.

Keep the construct short. Searching `responsible innovation` lets the dimensions emerge from the literature; adding `anticipation` presupposes the answer. Selecting a topic first is also worthwhile — the topic filter removes most cross-disciplinary noise.

After retrieval the tool reports what share of results actually mention the query in title or abstract. Below 60% it refuses the batch outright. This guards against a specific failure: a loose query combined with citation sorting returns famous papers that merely happen to contain the words somewhere in their full text.

Step 2 — draft

依 30 篇檢索到的文獻草擬維度

極性模型 ⓘ

☐ 由模型依文獻判斷 ☒ 指定使用雙極模型 ☐ 不使用極性

補充指示 (選填)

例如：維度請停留在個人實踐層次，不要拉到制度政策層次

草擬框架

草稿完成：3 個維度，均以檢索到的文獻為依據。

警告

- model's own coverage warning: The literature treats anticipation primarily as a single dimension of the broader Responsible Research and Innovation (RRI) framework; these dimensions are derived from the functional descriptions of anticipation within the provided texts rather than a formal sub-classification.
- 23 of 30 retrieved works were not used by any dimension

Figure 46. Retrieved abstracts are given to the model, which drafts dimensions. The warnings panel is the hallucination guard's output.

What the hallucination guard blocks

Every reference the model cites must exist in the retrieved set. Invented citations are removed; if all citations in a draft are invented the draft fails outright rather than being partially salvaged. Each dimension must also carry at least one grounding reference or it is discarded.

The message that 23 of 30 retrieved works went unused is not an error. It reports that seventy percent of the retrieval did not inform any dimension — possibly the query was too broad, possibly the model overlooked something, and either way it is worth checking yourself.

Step 3 — approve

Approve each dimension

Nothing enters the framework without an explicit decision. A dimension left undecided is treated as rejected: silence is not consent. Your decisions, edits and reasons are written into the framework file and can be reported in the methods section.

Anticipation · ANT · 1★ + 9

The ability to imagine and judge the potential future effects, unknown impacts, and alternative trajectories of an innovation. It involves using foresighting to govern in the present to shape uncertain futures.

Grounded in

★ Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. *Research Policy*, 42(9), 1568-1580. <https://doi.org/10.1016/j.respol.2013.05.008>

What the model says this work contributes: W1 establishes anticipation as one of the four integrated dimensions of responsible innovation.

Decision

☐ Not yet decided ☒ Approve as drafted ☐ Approve with edits ☐ Reject

Reason / note

Reflexivity · REF · 1★ + 6

The process of critical reflection on the goals, assumptions, and moral responsibilities inherent in the innovation process. It involves questioning the trajectory of novelty and identifying potential 'moral lock-ins'.

Grounded in

Figure 47. Each dimension is presented with its definition, its grounding references (★) and the model's account of what each reference contributes.

A dimension with no explicit decision does not enter the framework. Silence counts as rejection, not consent. Saving also requires a named approver: approval must be attributable.

想像、繪製或規劃替代性的未來路徑與軌跡，以了解技術如何塑造或適應不確定的未來。

定義依據

★ Mulderman, K., et al. (2020). Four approaches to anticipatory climate governance: Different conceptions of the future and implications for the present. *Wiley Interdisciplinary Reviews Climate Change*, 11(6). <https://doi.org/10.1002/wcc.673>

★ Eastwood, C., Edwards, P., & Turner, J. (2021). Review: Anticipating alternative trajectories for responsible Agriculture 4.0 innovation in livestock systems. *animal*, 15, 100296-100296. <https://doi.org/10.1016/j.animal.2021.100296>

模型說明該文獻的貢獻 W19 emphasizes the use of approaches to anticipate future trajectories.

決定

☒ 尚未決定 ☐ 原案通過 ☐ 修改後通過 ☐ 退回

原案通過 0、修改後通過 0、退回 0、尚未決定 3

核可者 (你的姓名)

Hung Chi Chang

框架檔名

responsible_innovation_anticipation_fram

對整份框架的說明 (選填)

儲存已核可的框架

Figure 48. Approval tally, approver name and framework filename. Decisions, edits and stated reasons are all written into the framework file.

Works cited by this framework

Citation	Role	Dimension	DOI
Guston, D. H. (2014). Understanding 'anticipatory governance'. Social Studies of Science, 44(2), 2	grounding	Anticipation	10.1177/0306312713508669
Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Research Policy, 42(9), 1568-1580.	grounding	Anticipation, Reflexivity, Public Engagement	10.1016/j.respol.2013.05.008
von Schomberg, R. (2013). A vision of responsible research and innovation. In R. Owen, J. Bessar, & D. Guston (Eds.), The Handbook of Science and Technology Studies (3rd ed., pp. 979-1000). MIT Press.	grounding	Anticipation, Responsiveness	10.1002/9781118551424.ch3
Owen, R., Macnaghten, P., & Stilgoe, J. (2012). Responsible research and innovation: From science to innovation. In R. Owen, J. Bessar, & D. Guston (Eds.), The Handbook of Science and Technology Studies (3rd ed., pp. 979-1000). MIT Press.	grounding	Reflexivity, Public Engagement	10.1093/scipol/scs093
Barben, D., Fisher, E., Selin, C., & Guston, D. H. (2008). Anticipatory governance of nanotechnology: Foresight, engagement, and integration. In E. J. Hackett et al. (Eds.), The Handbook of Science and Technology Studies (3rd ed., pp. 979-1000). MIT Press.	grounding	Anticipation	
Stirling, A. (2008). 'Opening up' and 'closing down': Power, participation, and pluralism in the social sciences. Science and Technology Studies, 21(1), 1-10.	grounding	Public Engagement	10.1177/0162243907311265
Wilsdon, J., & Willis, R. (2004). See-through Science: Why Public Engagement Needs to Move Upside Down. Science and Technology Studies, 17(1), 1-10.	grounding	Public Engagement	
Pellizzoni, L. (2004). Responsibility and environmental governance. Environmental Politics, 13(3), 311-328.	grounding	Responsiveness	10.1080/0964401042000229034
Wynne, B. (1992). Misunderstood misunderstanding: Social identities and public uptake of science. Science and Technology Studies, 5(1), 1-10.	grounding	Reflexivity	10.1088/0963-6625/1/3/004
Muiderman, K., Gupta, A., Vervoort, J., & Biermann, F. (2020). Four approaches to anticipatory climate governance. Science and Technology Studies, 33(1), 1-10.	supporting	Anticipation	10.1002/wcc.673

Figure 49. The active framework lists every work it cites with role, dimension and DOI.

維度定義

▼ 預期 · ANT

系統性的前瞻：探索多重可能的未來、處理不確定性，並在技術定型之前保留設計彈性。

長期／探索式 (+)

- 情境規劃
- 上游介入
- 承認不確定性
- 迴避技術鎖定

短視／決定論 (-)

- 單一確定性預測
- 延後前瞻
- 把預期降格為 KPI 追蹤

定義依據

★ Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Research Policy, 42(9), 1568-1580.

★ Barben, D., Fisher, E., Selin, C., & Guston, D. H. (2008). Anticipatory governance of nanotechnology: Foresight, engagement, and integration. In E. J. Hackett et al. (Eds.), The Handbook of Science and Technology Studies (3rd ed., pp. 979-1000). MIT Press.

★ Guston, D. H. (2014). Understanding 'anticipatory governance'. Social Studies of Science, 44(2), 218-242.

Figure 50. Dimension definitions with positive and negative indicators and grounding literature. Each dimension of the built-in responsible innovation framework carries six to seven references, nearly all with DOIs.

Why the provenance record matters

"Dimensions an engineer chose from memory" and "dimensions retrieved systematically, ranked by citation, with recorded inclusion and exclusion and per-dimension named approval" are different objects under review. The active framework page also generates a methods paragraph whose figures come from the recorded construction process.

10. Export

RI-AutoCoder PRO

Responsible Innovation (AIRR) · Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. Research Policy, 42(9), 1568-1580.

Run analysisData & descriptorsCode reviewCross-analysisTheme structureLexicon inductionReliabilityExportFramework builder

Download results

The Excel file contains the coded long table, descriptors, one sheet per cross-analysis, and a code reference. The long table is tidy data and can be loaded straight into SPSS, R or NVivo.

Excel (full cross-analysis)Word report

Merged JSON

Figure 51. Excel carries the full set of cross-analysis tables, Word is an editable report draft, and JSON is the complete dataset, suitable as supplementary material.

11. Troubleshooting

Symptom	Cause and remedy
Failed to import encodings module at launch	The <code>.venv</code> folder was copied from elsewhere. Run the launcher again — it detects this and rebuilds — or delete <code>.venv</code> manually.
Model returns 404 or is not found	That model has been retired. The tool queries the API for the current list; re-entering the key refreshes it.
Every figure on the lexicon tab is zero	The lexicon does not match the active framework, or the file was not found. An explicit message appears on screen. Switch back to the matching framework, or build a lexicon for the current one.
Yellow warning at the top of the lexicon tab	No Chinese segmenter is installed and the statistical fallback is in use, so some entries will be wrong word boundaries. Install with <code>pip install ckip-transformers</code> and restart.
Chi-square reports no p-value	Deliberate. Expected counts are too low for the test's assumptions to hold.
Cohen's κ is low but agreement is high	The kappa paradox, arising from sparse marking rather than coder disagreement. Report all three coefficients and explain.
Quota exhausted mid-analysis	Swap keys, keep the same selection, press Start again. Completed files are skipped and never billed twice.
The reliability tab asks for transcripts again	

Only records saved by older versions need this. Transcripts analysed by this tool are carried automatically.

12. Honest limitations

A credible description of a tool has to state what it cannot do.

- **Model coding is not ground truth.** It is a draft. Coding that has not been human-reviewed should not be written up as verified.
- **Lexicon quality depends on the segmenter.** Without CKIP the statistical fallback has no linguistic knowledge of Chinese and will produce wrong boundaries. Semantic coding does not pass through segmentation and is unaffected.
- **Many statistics are meaningless at small n.** The tool withholds figures whose assumptions fail, but it cannot tell you whether six respondents support your conclusion. That is a research design question.
- **Framework construction depends on OpenAlex coverage.** Niche fields, non-English literature and very recent work may not be retrievable. The retrieval log records honestly how much was actually found.
- **This tool will not make a study rigorous.** It makes what you did traceable, reportable and checkable.

13. References

The built-in responsible innovation framework rests on:

- Stilgoe, J., Owen, R., & Macnaghten, P. (2013). Developing a framework for responsible innovation. *Research Policy*, 42(9), 1568–1580. <https://doi.org/10.1016/j.respol.2013.05.008>
- Owen, R., Macnaghten, P., & Stilgoe, J. (2012). Responsible research and innovation. *Science and Public Policy*, 39(6), 751–760. <https://doi.org/10.1093/scipol/scs093>
- von Schomberg, R. (2013). A vision of responsible research and innovation. In *Responsible Innovation* (pp. 51–74). Wiley.

Statistical methods:

- Monroe, B. L., Colaresi, M. P., & Quinn, K. M. (2008). Fightin' words. *Political Analysis*, 16(4), 372–403.
- Gwet, K. L. (2008). Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1), 29–48.
- Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research. *Organizational Research Methods*, 16(1), 15–31.